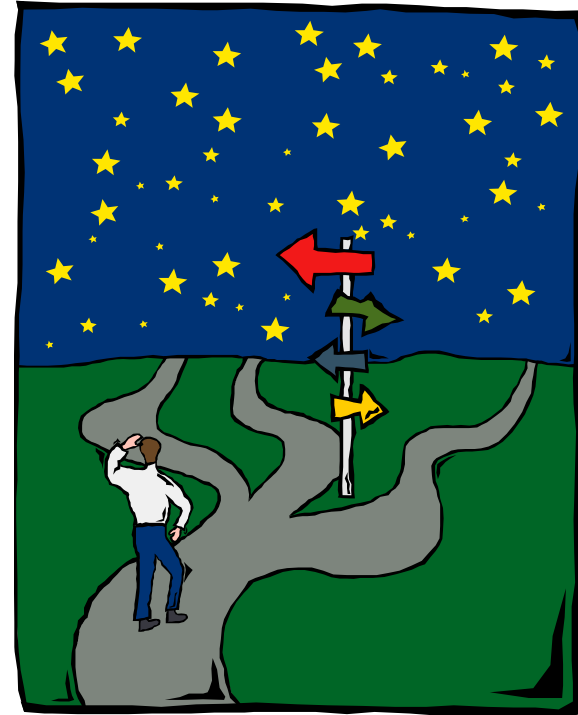


Single View Geometry

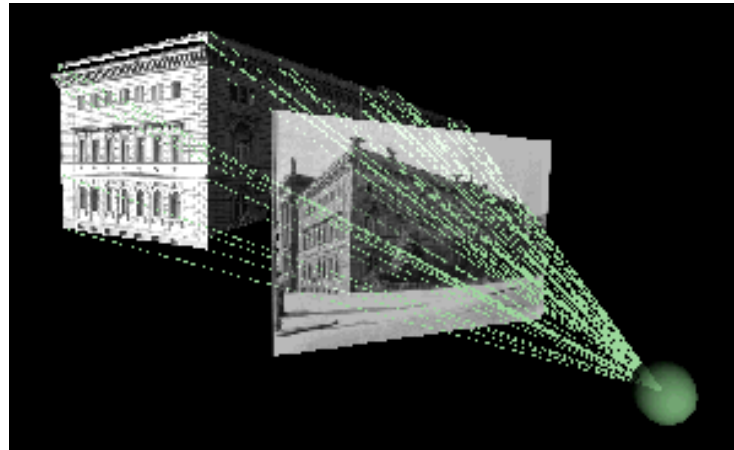
**Camera model &
Orientation +
Position estimation**

Jianbo Shi



What am I?

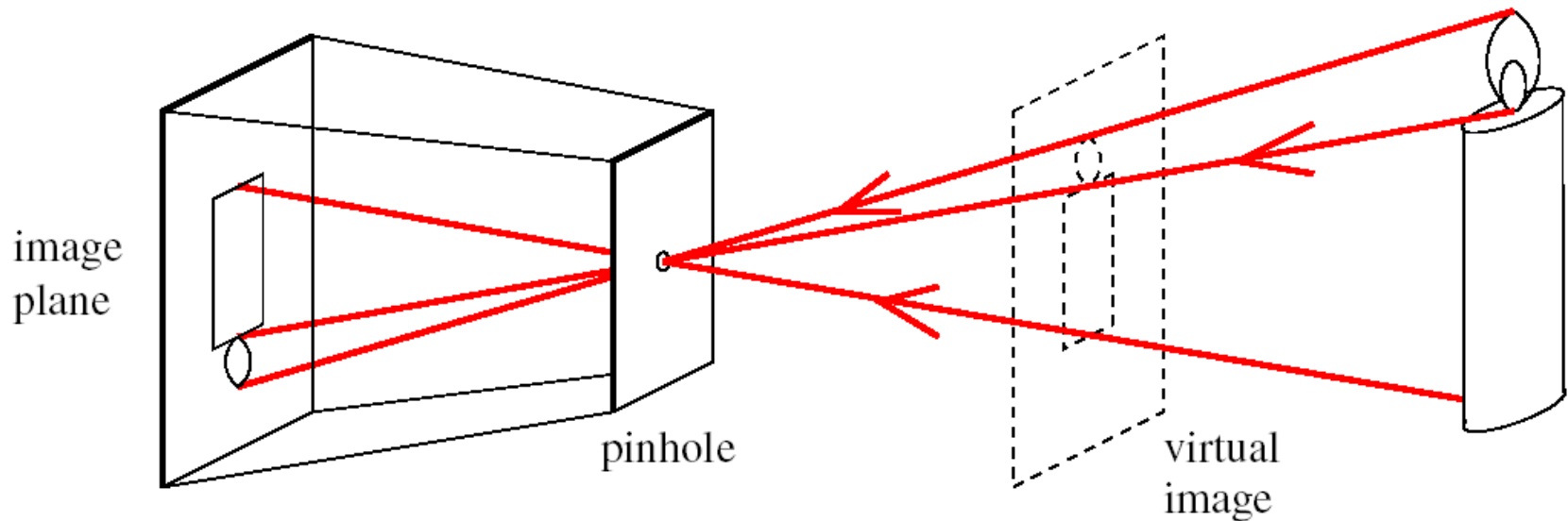
Camera projection model



The overall goal is to compute 3D geometry of the scene from just 2D images. We will first study 3D->2D camera projection using linear algebra.

The Pinhole Camera

- Light enters a darkened chamber through a pinhole opening and forms an image on the further surface



The Pinhole Camera Model

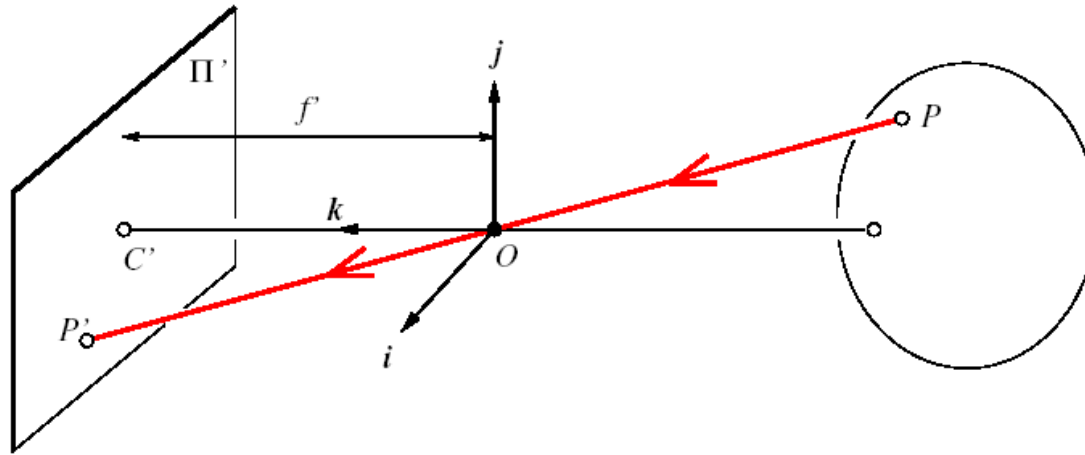


Figure 1.4. The perspective projection equations are derived in this section from the colinearity of the point P , its image P' and the pinhole O .

Let P denote a scene point with coordinates (x, y, z) and P' denote its image with coordinates (x', y', z') . Since P' lies in the image plane, we have $z' = f'$. Since the three points P , O and P' are colinear, we have $\overrightarrow{OP'} = \lambda \overrightarrow{OP}$ for some number λ , so

$$\begin{cases} x' = \lambda x \\ y' = \lambda y \\ f' = \lambda z \end{cases} \iff \lambda = \frac{x'}{x} = \frac{y'}{y} = \frac{f'}{z},$$

and therefore

$$\begin{cases} x' = f' \frac{x}{z}, \\ y' = f' \frac{y}{z}. \end{cases}$$

(1.1.1)

Ideal case:

Projection equation:

$$x' = f X / Z$$

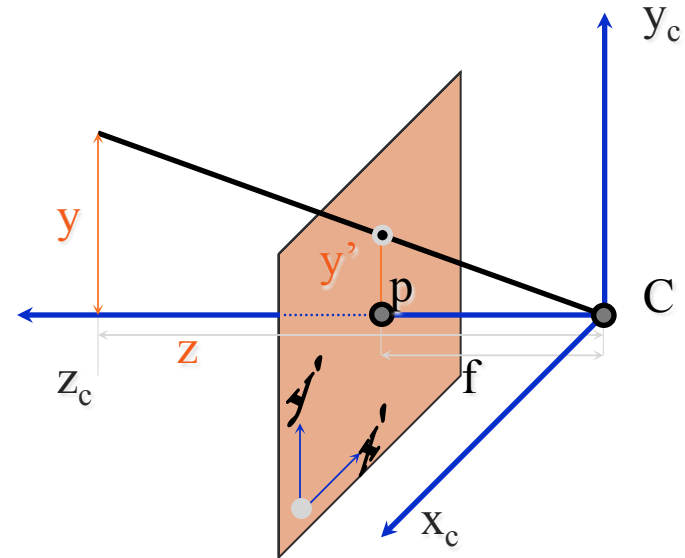
$$y' = f Y / Z$$



$$Zx' = f X$$

$$Zy' = f Y$$

$$Z = Z$$



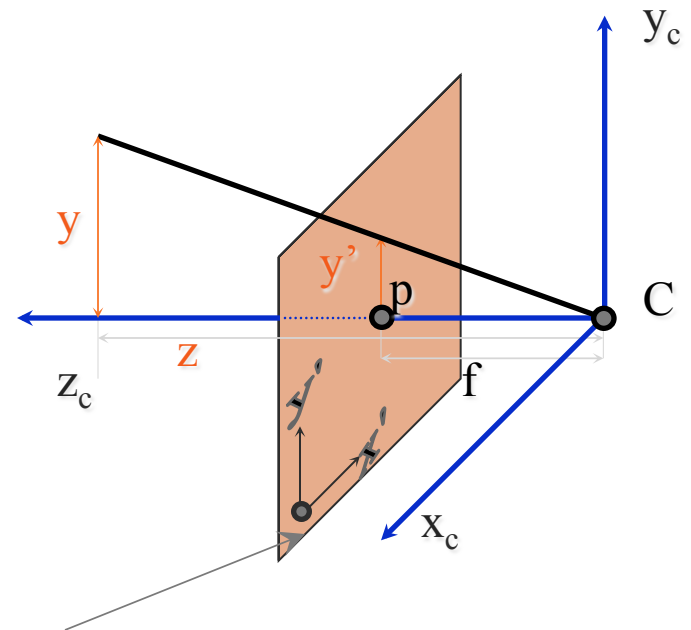
Step 1: Camera projection matrix

$$Zx' = fX$$

$$Zy' = fY$$

$$Z = Z$$

$$\begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} Z_c$$

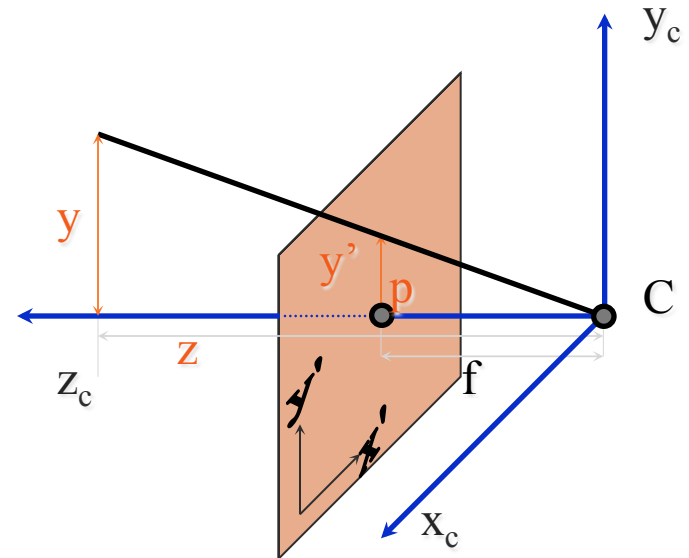


$$\begin{matrix} \nearrow & \uparrow & \uparrow \\ \mathbf{P}_0 & \mathbf{X} & = \mathbf{x} \end{matrix}$$

Step 2: Intrinsic camera parameters: map camera coordinate to pixel coordinate

$$\begin{bmatrix} \alpha_x & s & p_x \\ 0 & \alpha_y & p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$\mathbf{K}$ (3x3 submatrix) Optical world Pixel world



α_x, α_y is pixel scaling factor

p_x, p_y is the principle point (where optical axis hits image plane)

s is the slant factor, when the image plane is not normal to the optical axis

Combine the Intrinsic camera parameters

$$\begin{bmatrix} \alpha_x & s & p_x \\ 0 & \alpha_y & p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} \alpha_x f & s & p_x & 0 \\ 0 & \alpha_y f & p_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

K

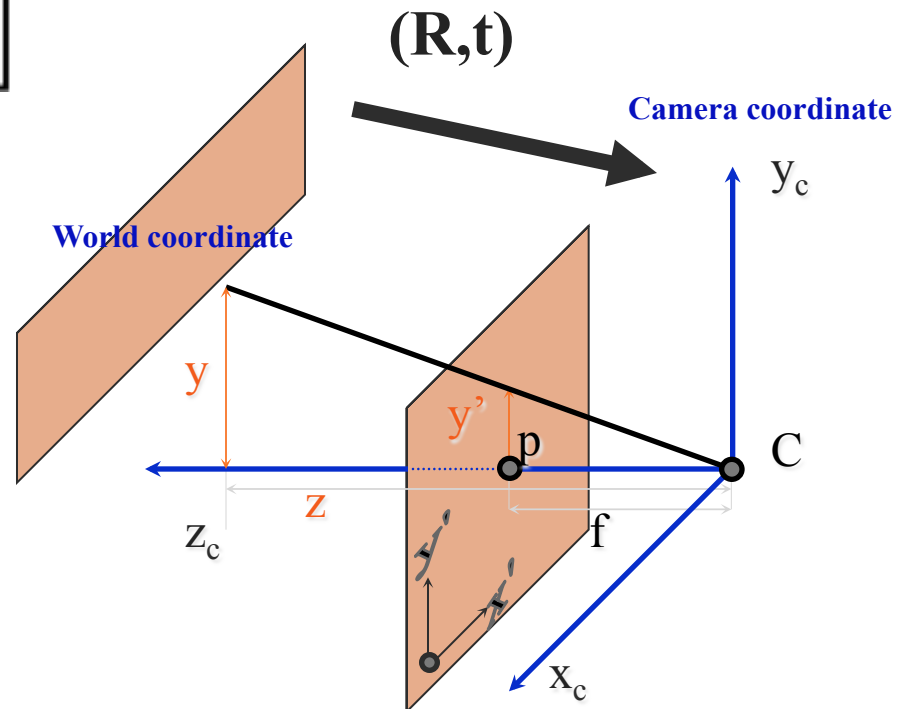
(Calibration matrix)

$$\mathbf{P}_0 \mathbf{X} = \mathbf{x}$$

$$\mathbf{P} = [\mathbf{K}, \mathbf{0}] = \mathbf{K} [\mathbf{I}, \mathbf{0}]$$

Step 3: External parameters: rotation and translation map the world to camera coordinates

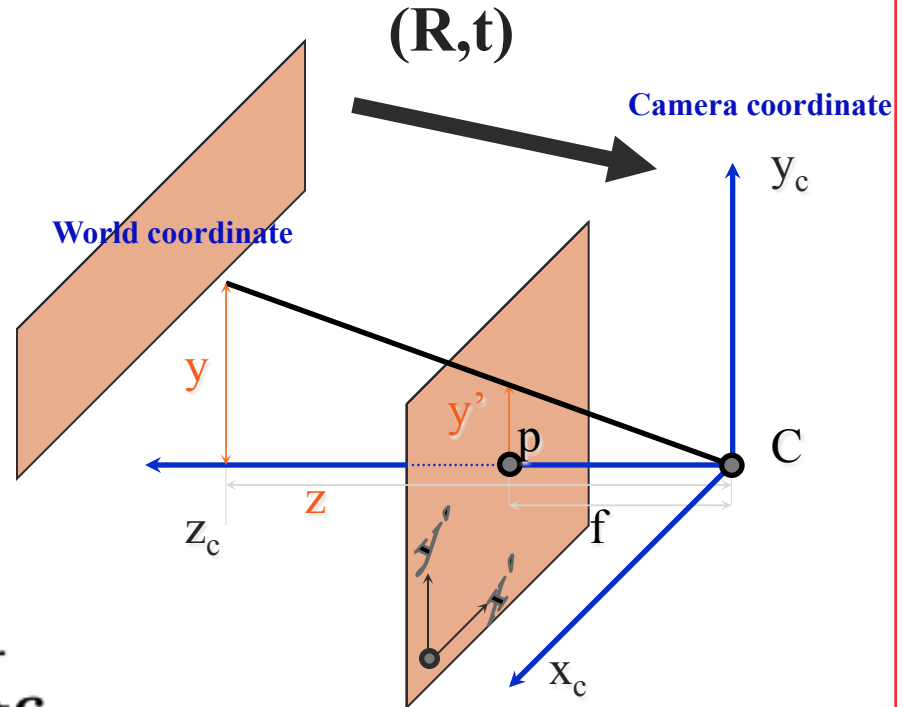
$$\mathbf{X}_c = \begin{bmatrix} \mathbf{R}_{3 \times 3} & \mathbf{t}_{3 \times 1} \\ \mathbf{0} & 1 \end{bmatrix} \mathbf{X}$$



Combining Internal and External parameters

$$1) \quad \mathbf{X}_c = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \mathbf{X}$$

$$2) \quad \mathbf{x} = \mathbf{K}_{3 \times 3} [\mathbf{I}; 0]_{3 \times 4} \mathbf{X}_c$$



- 1) Translate the world coordinate into the camera coordinate
- 2) Translate the camera coordinate into the pixel coordinate

Combining Internal and External parameters

$$\mathbf{X}_c = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \mathbf{X}$$

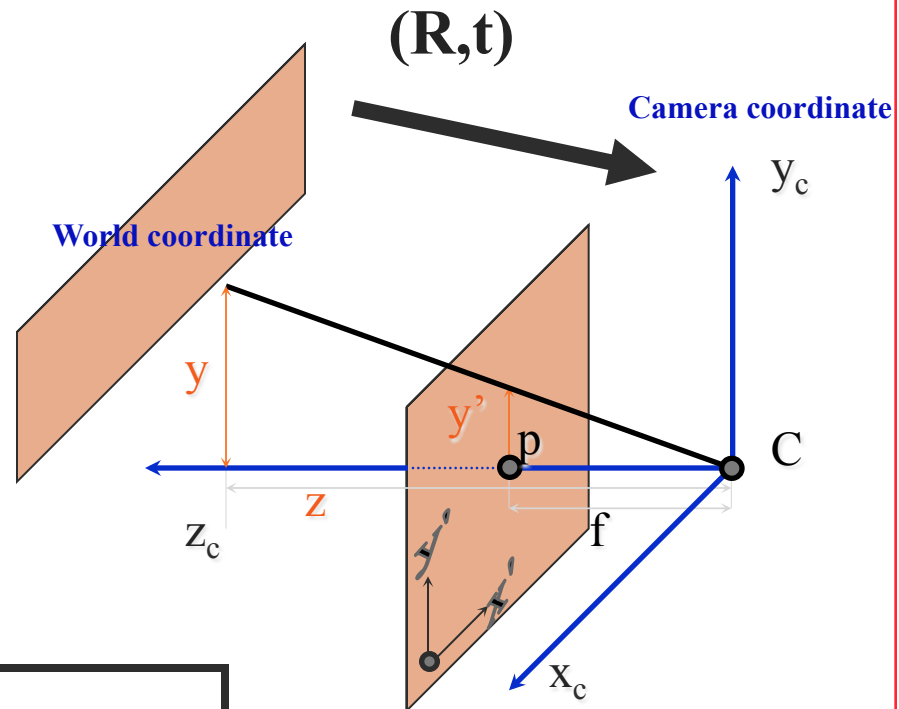
$$\mathbf{x} = \mathbf{K}_{3 \times 3} [\mathbf{I}; 0]_{3 \times 4} \mathbf{X}_c$$

After simplification:

$$\mathbf{x} = \mathbf{K} [\mathbf{R}, \mathbf{t}] \mathbf{X}$$

pixel

world

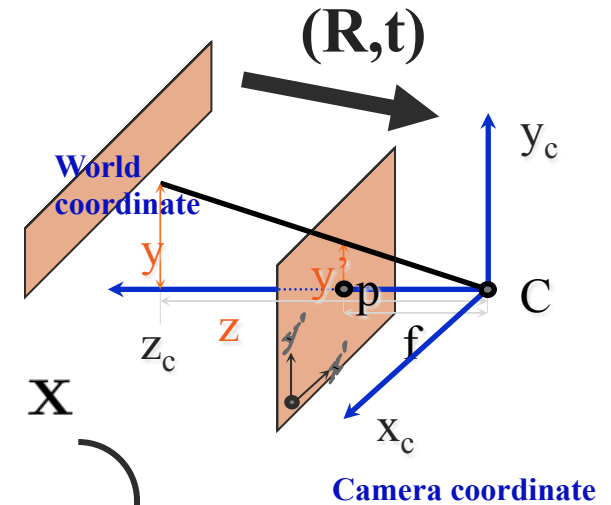


Special case, planar world, homograph

$$X = (x_w, y_w, \boxed{z_w=0}, 1)$$

Expand: $x = K [R, t] X$

$$x = K \begin{pmatrix} r1 & r2 & r3 \\ t \end{pmatrix} \begin{pmatrix} x_w \\ y_w \\ \cancel{z_w=0} \\ 1 \end{pmatrix}$$



Special case, planar world, homograph

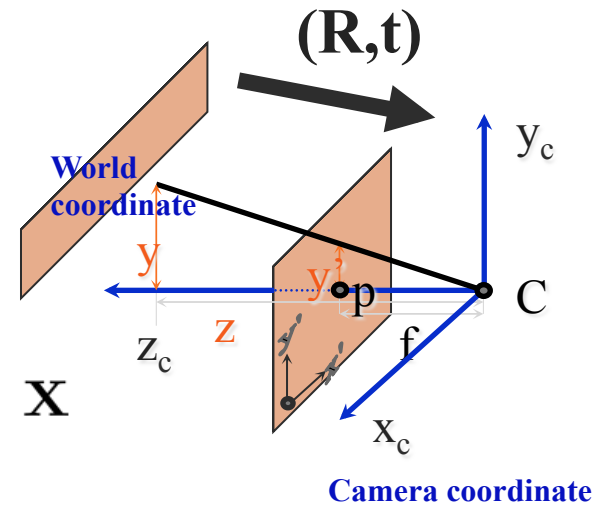
$$X = (x_w, y_w, \boxed{z_w=0}, 1)$$

After simplification:

$$x = K \begin{pmatrix} r1 & r2 & t \end{pmatrix} \begin{pmatrix} x_w \\ y_w \\ 1 \end{pmatrix}$$

We have the homographic mapping:

$$x = K H_{3 \times 3} X$$



Special case: rotating camera

Expand: $\mathbf{x} = \mathbf{K} [\mathbf{R}, \mathbf{t}] \mathbf{X}$

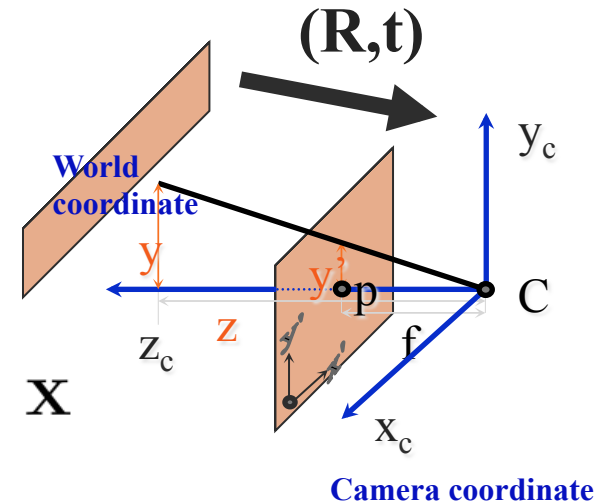
with $\mathbf{t} = 0$

$$\mathbf{x} = \mathbf{K} [\mathbf{R}] \mathbf{X}$$

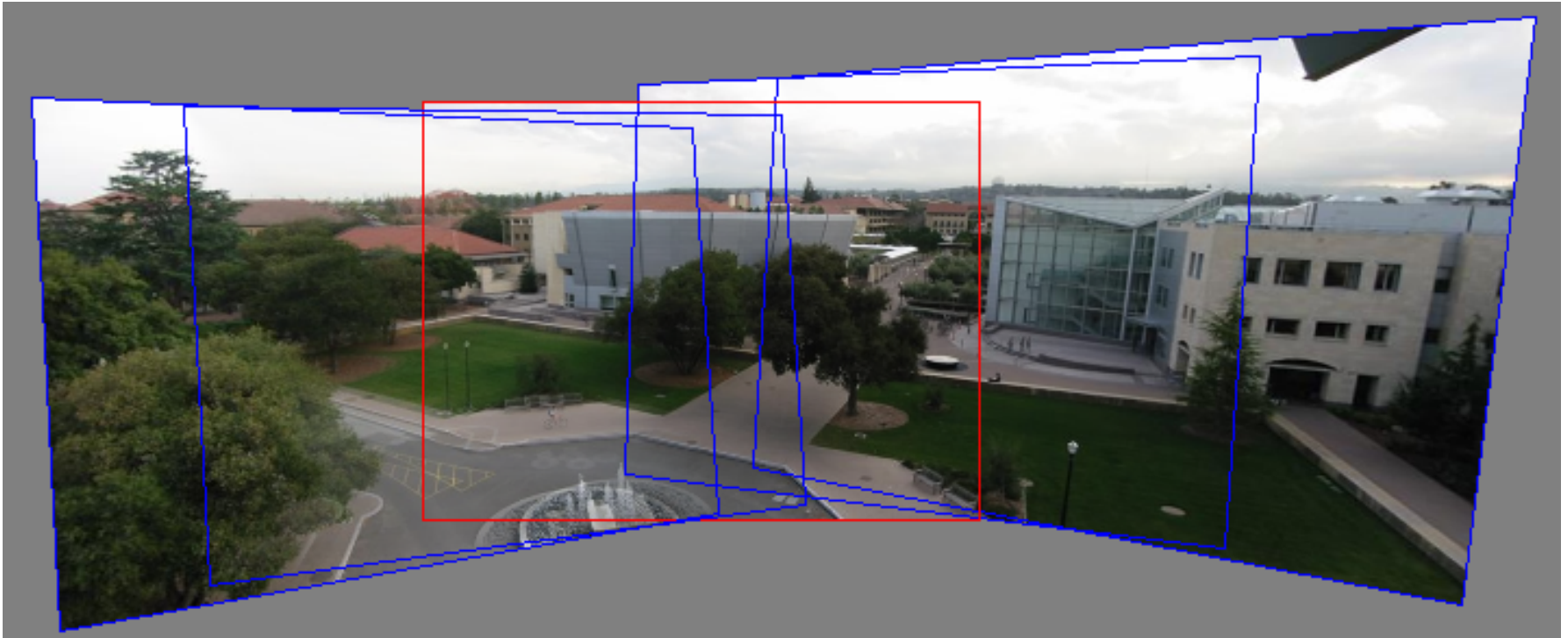
$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_w \\ \mathbf{y}_w \\ \mathbf{Z}_w \end{bmatrix}$$

This is also a homography with

$$\mathbf{H} = \mathbf{R}$$



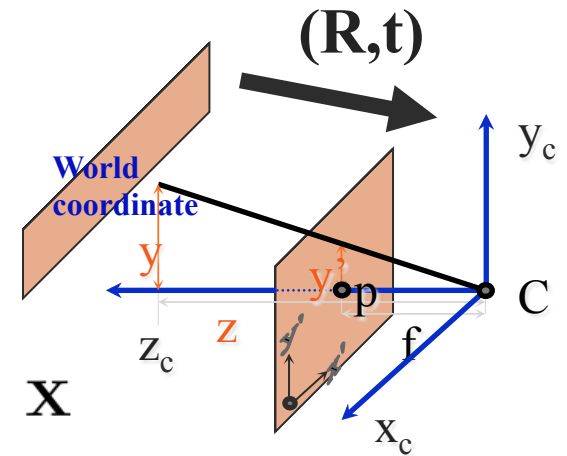
Panoramas



$$\mathbf{x} = \mathbf{K} [\mathbf{R}] \mathbf{X}$$

Visual Odometry: Overview

- 1) The transformation $(\mathbf{R}, \mathbf{t})$ tells us how to rotate + move the world coordinate to the camera coordinate.
- 2) In the following slides, we will see
 - 1) how to get the world reference's frame orientation+position from $(\mathbf{R}, \mathbf{t})$,
 - 2) how to get the pan/tilt/yaw angles from $\mathbf{R}$
- 3) If we want to know the camera orientation+position, we need to look at its inverse transform, which we should see is $(R^T, R^T(-t))$,
 - 1) The computation of the camera pan/tilt/yaw angle is the same as in the previous case

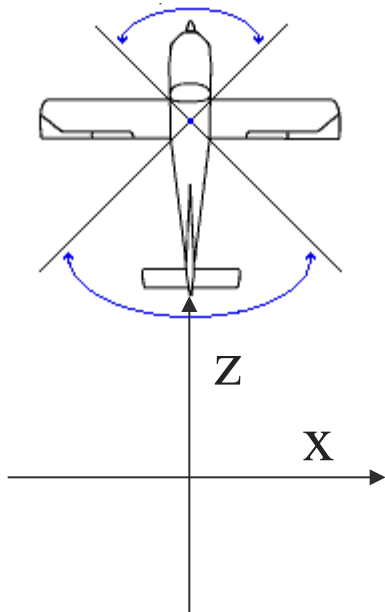


Recover **Camera** orientation

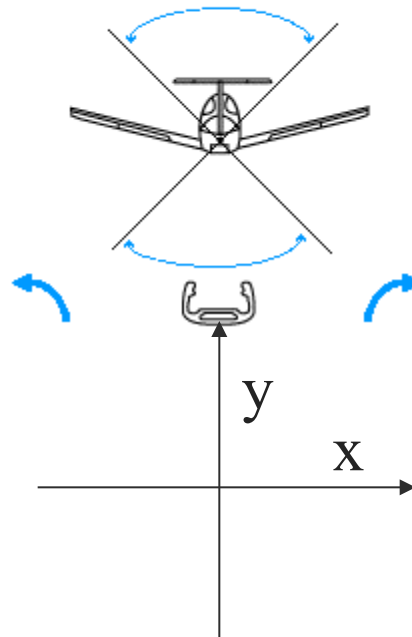
Case 1: single vanishing point (z-direction)

Pan/tilt/yaw angles

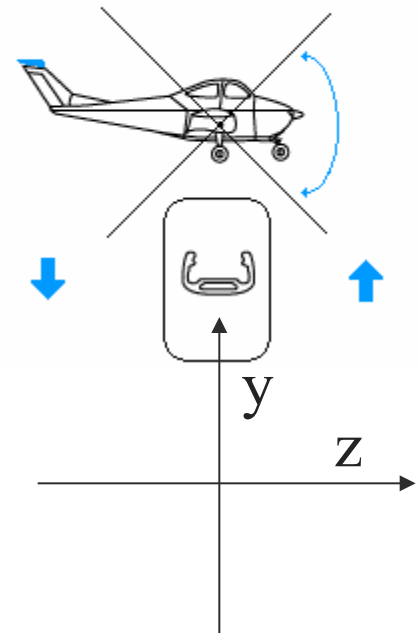
Pan α : rotation
around y-axis



Yaw γ : rotation
around z-axis

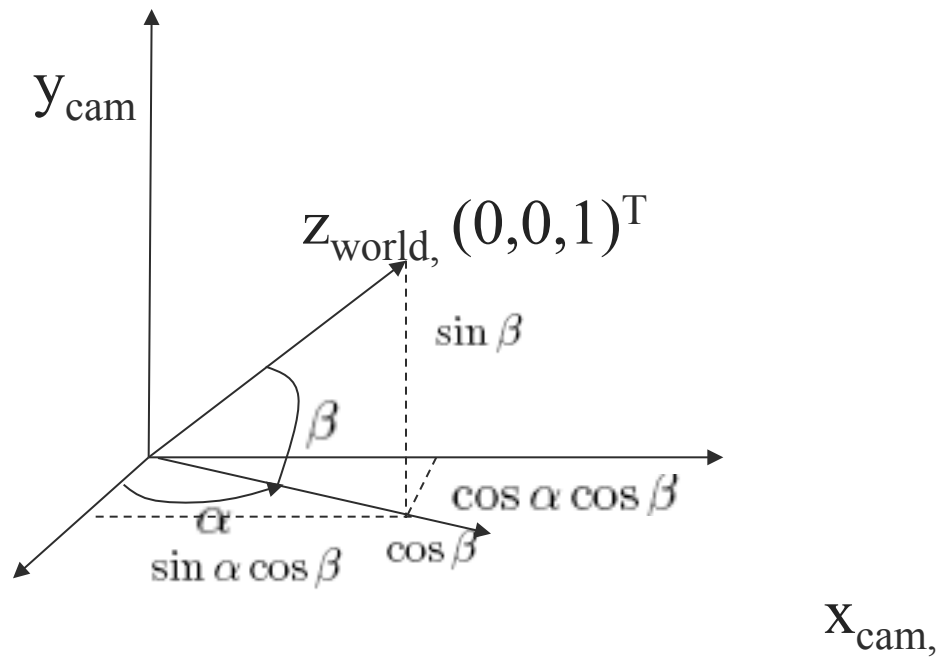


Tilt β : rotation
around x-axis

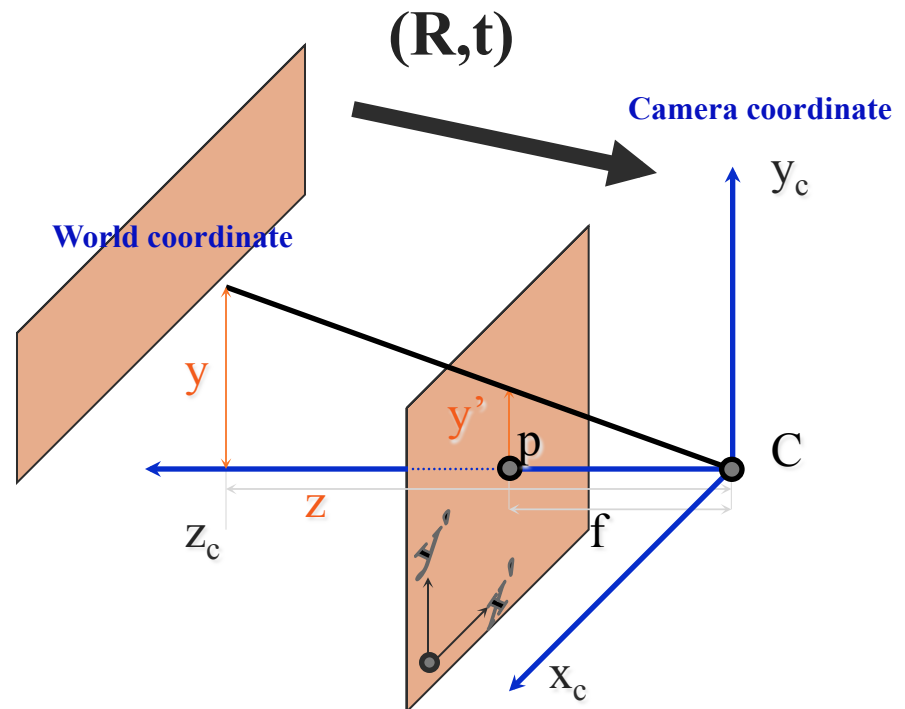
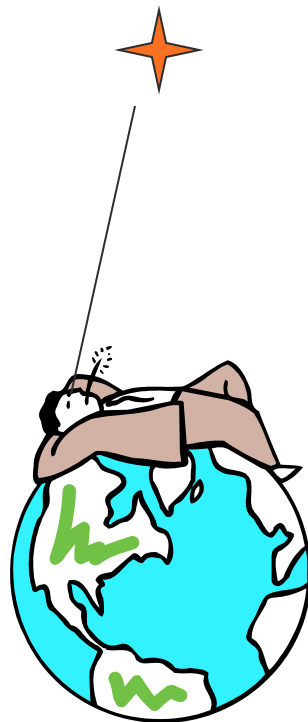


Rotation: Pan/Tilt/Yaw

Given the 3x3 rotation matrix R , we can recover, pan/tilt/yaw angle, and vise. vica.



The basic idea is that if we can see the “north” star,
we know how to oriented ourselves (minus yaw angle) (why?)
And the “north” star is just the point of infinity in some direction



Seeing the vanishing point can tell us:

Camera projection equation: $\mathbf{x} = \mathbf{K} \cdot [\mathbf{R}, \mathbf{t}] \cdot \mathbf{X}$, or

$$\mathbf{K} \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \mathbf{R} & \cdot & \mathbf{t} \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ t_w \end{bmatrix} = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} \quad [\mathbf{R}, \mathbf{t}] = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 & \mathbf{t} \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

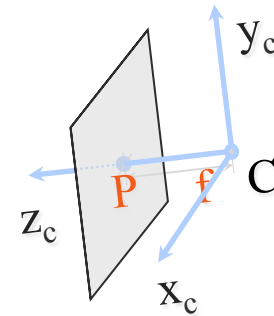
Columns of R matrix = image of vanishing points of world-space axes!

$\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3$ == image of x,y,z axis vanishing points

$$\begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 & \mathbf{t} \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{r}_3 \text{ (need to normalize to norm = 1)}$$

z-direction

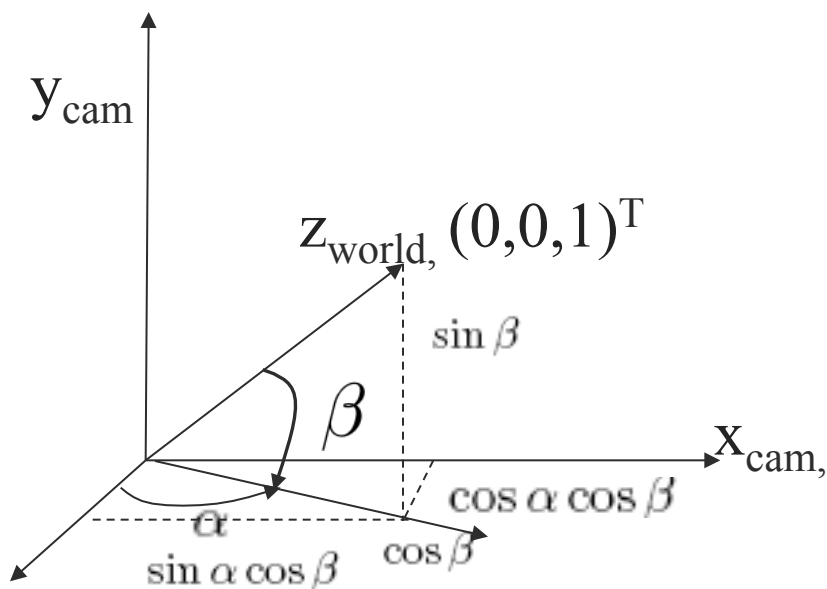
vanishing point



Recover Pan/Tilt angle from z-vanishing point

Geometric explanation: Pan α Tilt β

The alignment between z-axis of the world and image is defined only by the pan/tilt angle:

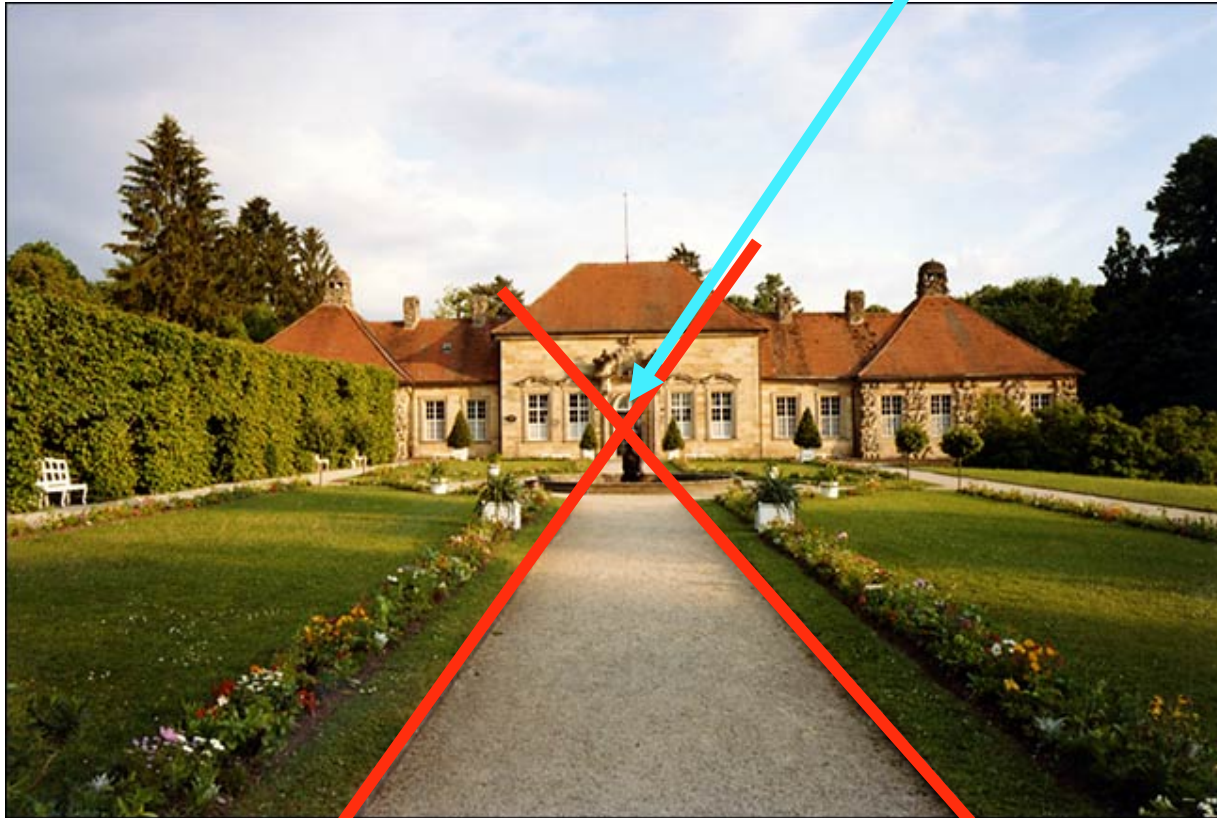


$$\mathbf{r}_3 = \begin{bmatrix} \sin \alpha \cos \beta \\ \sin \beta \\ \cos \alpha \cos \beta \end{bmatrix}$$

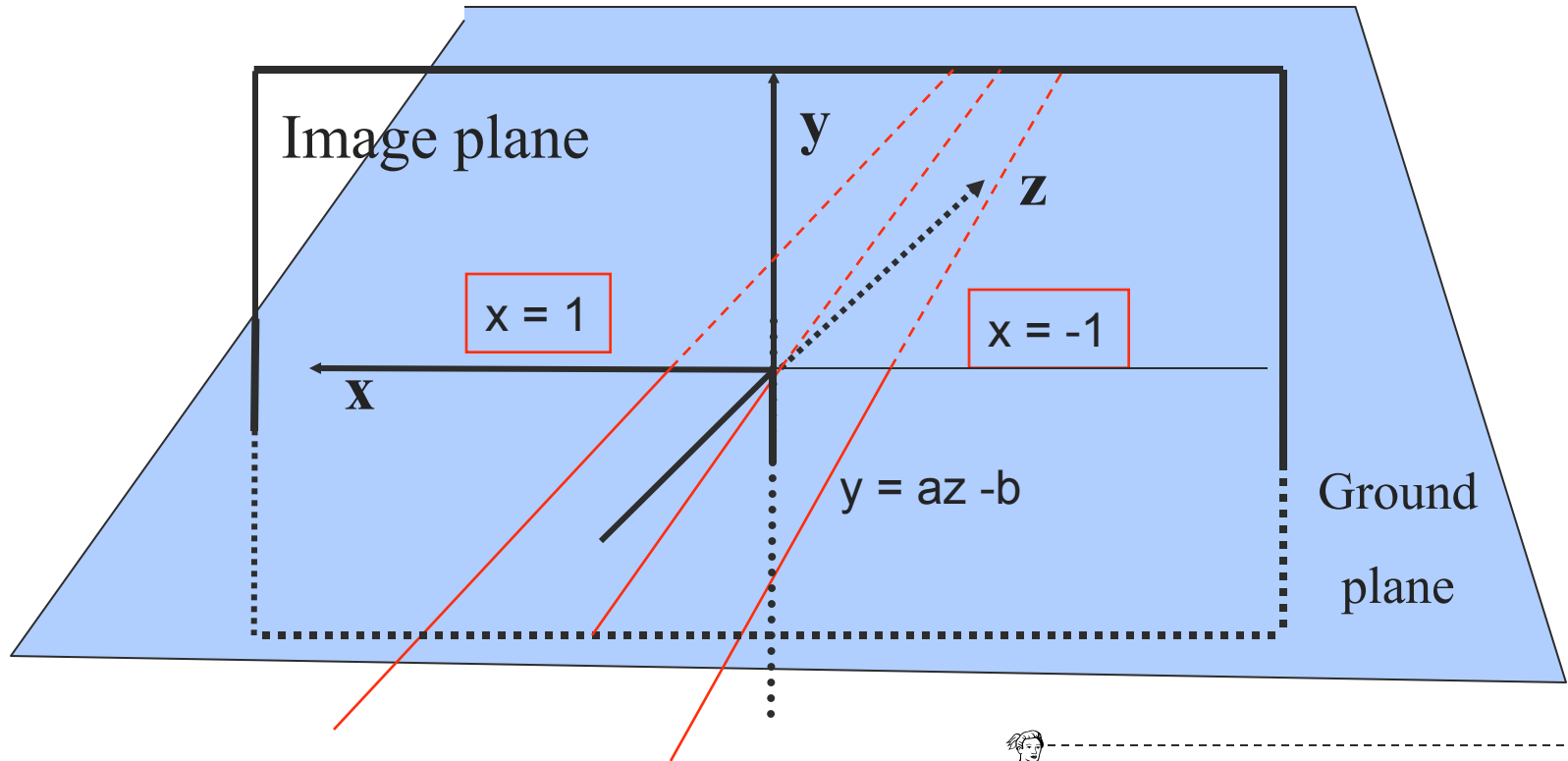
Note: tilt angle with rotation around x-axis is positive when pointing down in this figure

Sanity check

Vanishing point in z

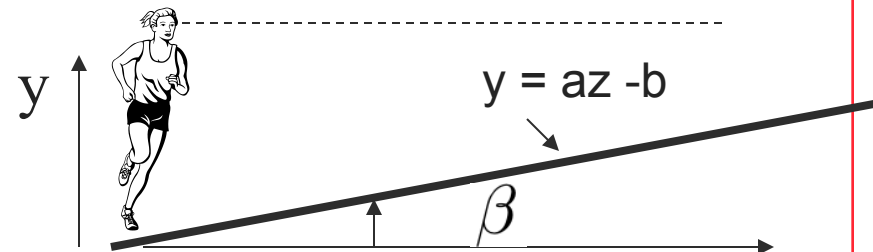


In this example, we are walking down a track, the ground is tilted up (we are going up the hill). Assume we are b meter tall. We see two parallel lines to our left and right, with $x = 1, -1$. A line on the ground is described by equation $y = az - b$, where a is the slope the ground.



So we know: $\tan(\beta) = a$

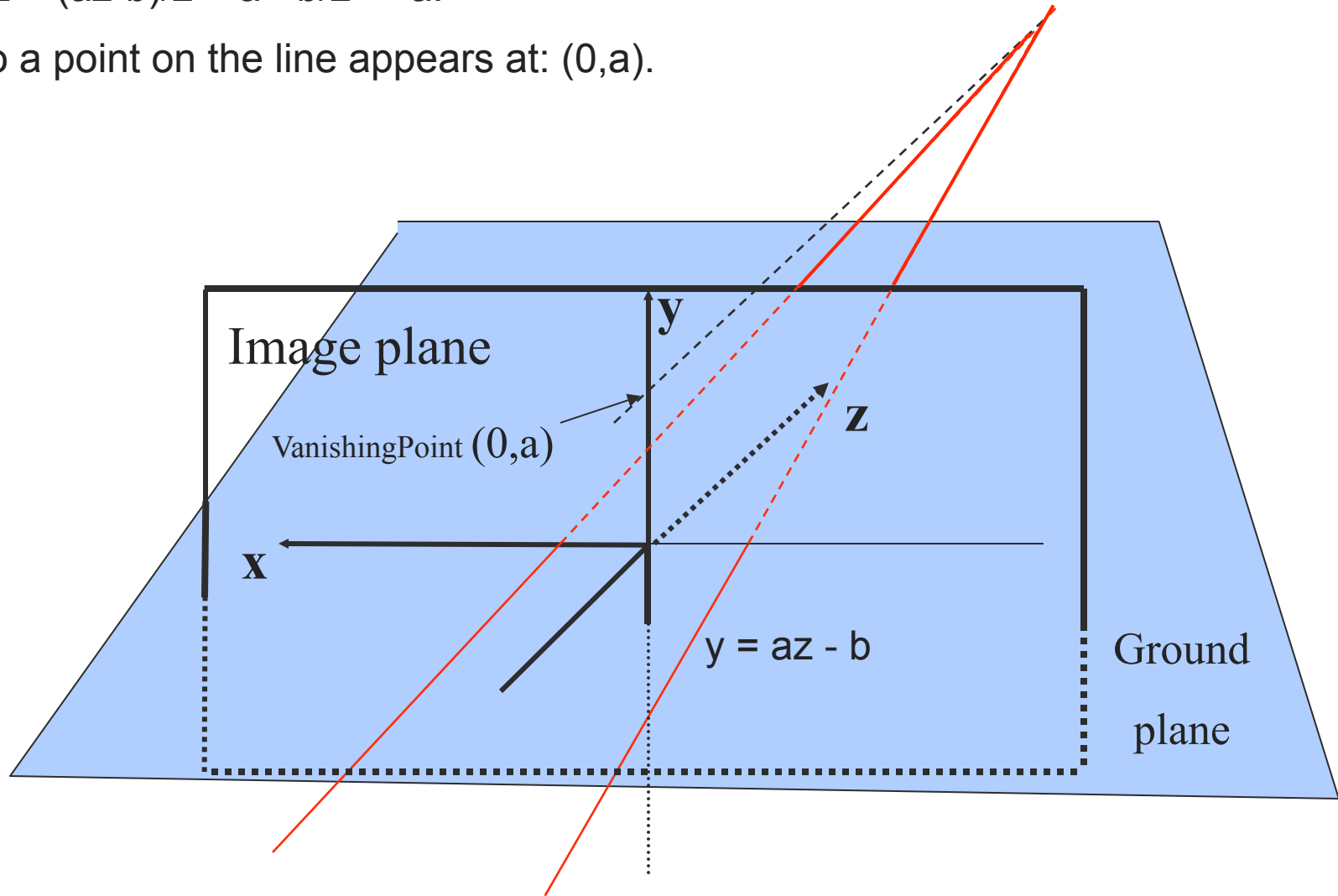
Tilt angle

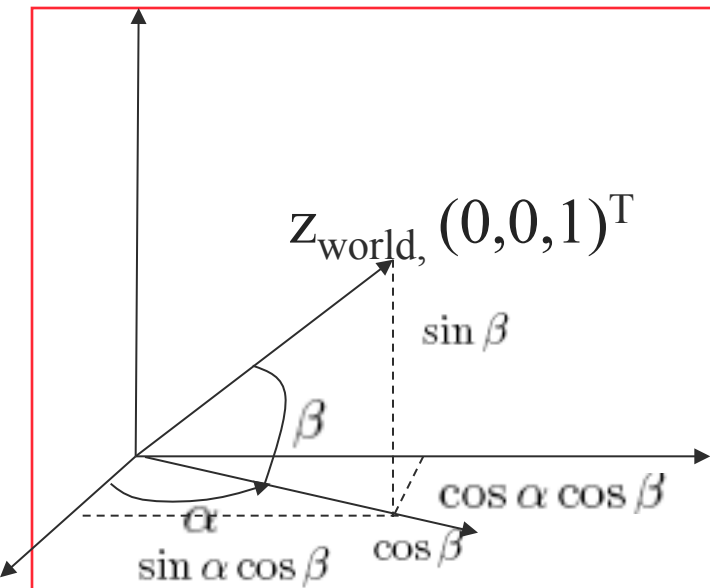


As a line gets far away, $z \rightarrow \text{infinity}$. If $(1,y,z)$ is a point on this line, its image is $f(1/z, y/z)$. As $z \rightarrow \text{infinity}$, $1/z \rightarrow 0$. What about y/z ?

$$y/z = (az-b)/z = a - b/z \rightarrow a.$$

So a point on the line appears at: $(0,a)$.

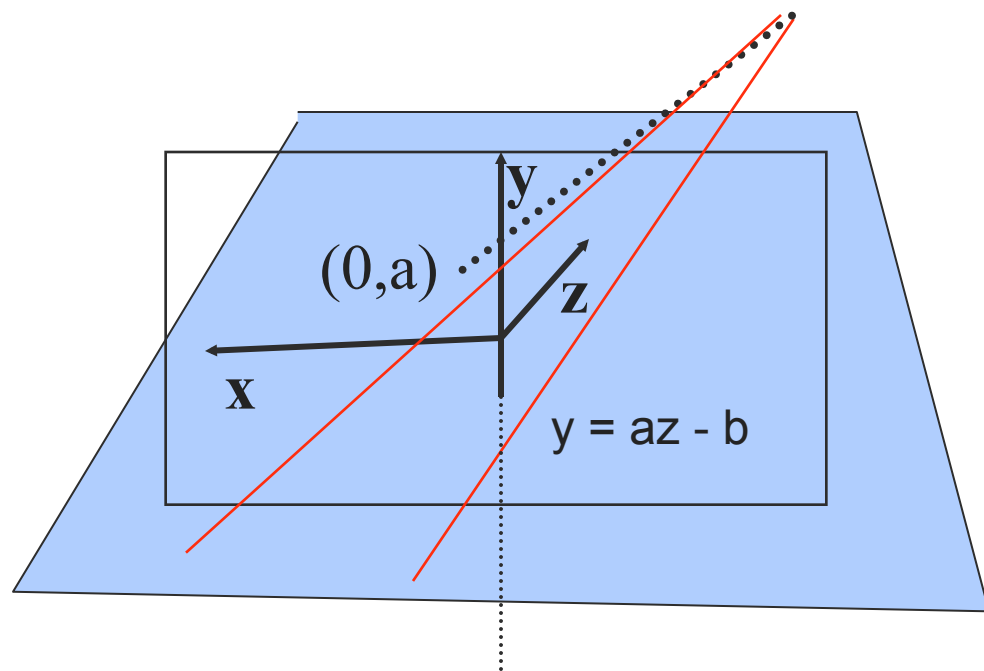




$$\mathbf{r}_z = \begin{bmatrix} \sin \alpha \cos \beta \\ \sin \beta \\ \cos \alpha \cos \beta \end{bmatrix} = \begin{bmatrix} 0 \\ a/\sqrt{1+a^2} \\ 1/\sqrt{1+a^2} \end{bmatrix}$$

Now, we check

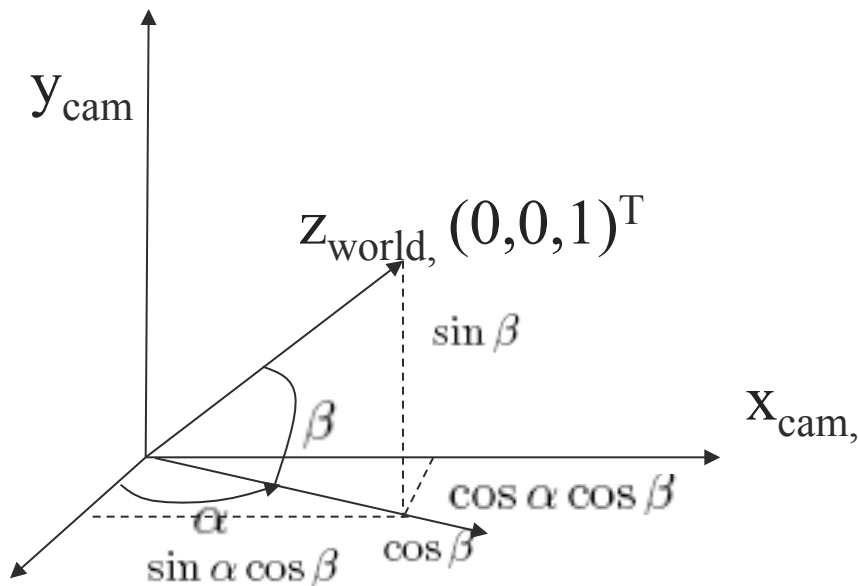
$$\rightarrow \tan(\beta) = a$$



Recover Pan/Tilt angle from z-vanishing point

Geometric explanation: Pan α Tilt β

The alignment between z-axis of the world and image is defined only by the pan/tilt angle:



$$\mathbf{r}_3 = \begin{bmatrix} \sin \alpha \cos \beta \\ \sin \beta \\ \cos \alpha \cos \beta \end{bmatrix}$$

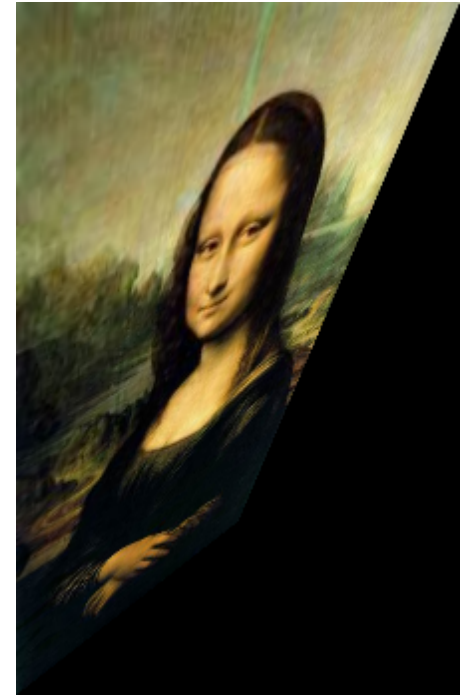
$$\alpha = \tan^{-1}(\mathbf{r}_3(1)/\mathbf{r}_3(3))$$

$$\beta = \sin^{-1}(\mathbf{r}_3(2))$$

Recover Camera orientation

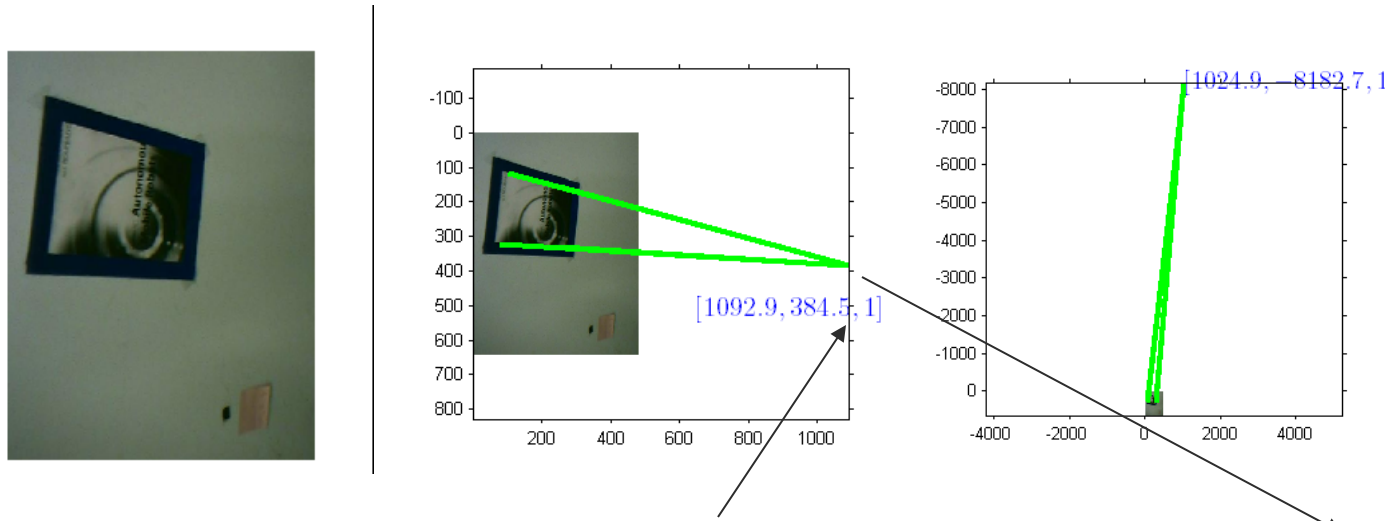
Case 2: two vanishing points, x-y direction

In some case, we can't measure the z-vanishing point directly, but if we have x-y vanishing points, we can imagine what the z-vanishing point



Basic idea:

- 1) x-y vanishing point gives us the first two columns of $\mathbf{R}$, $\mathbf{r}_1$ and $\mathbf{r}_2$
- 2) z-vanishing point, the third columns of $\mathbf{R}$, can be computed by $\mathbf{r}_3 = \mathbf{r}_1 \times \mathbf{r}_2$



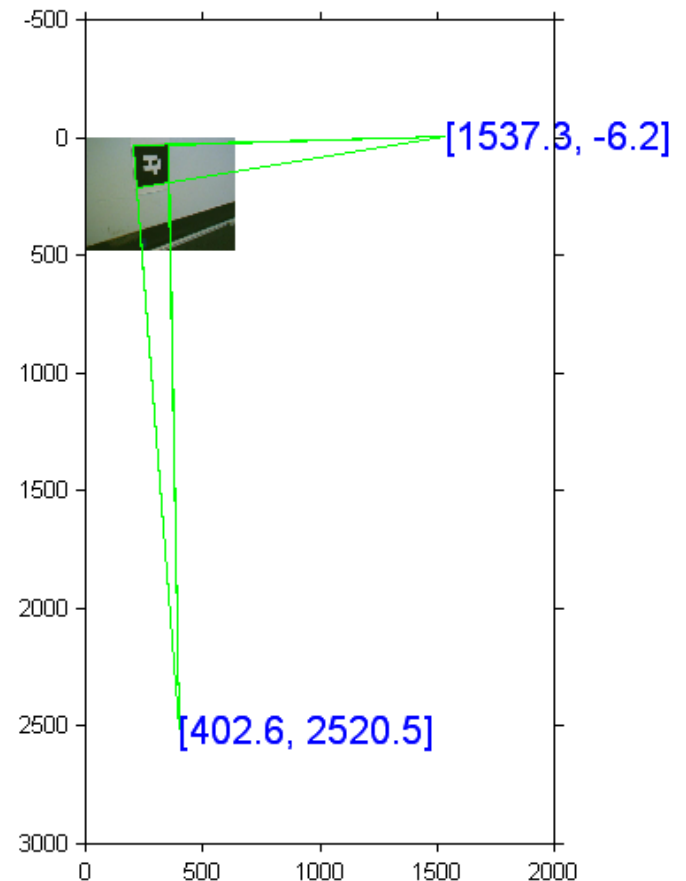
Recall: vanishing points of x,y direction, give us the, $\mathbf{r}_1$ and $\mathbf{r}_2$

$$\begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 & \mathbf{t} \end{bmatrix}$$

first two columns of $\mathbf{R}$

Steps for Recovering R from Two vanishing Point

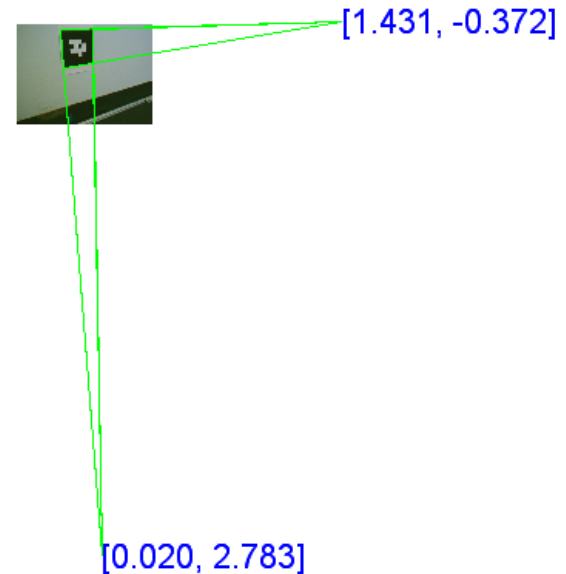
1. Measure X / Y direction vanishing p



Recovering R from Two vanishing Point (2)

2. Translate to the camera coordinates by intrinsic matrix K

$$x_{cam} \sim K^{-1} \cdot \begin{pmatrix} x_{img,x} \\ x_{img,y} \\ 1 \end{pmatrix}$$



Recovering R from Two vanishing Point (3)

3. Get r_1 and r_2 from vanishing point x_{vX} and x_{vY}

$$r_1 = x_{vX} / \|x_{vX}\|$$

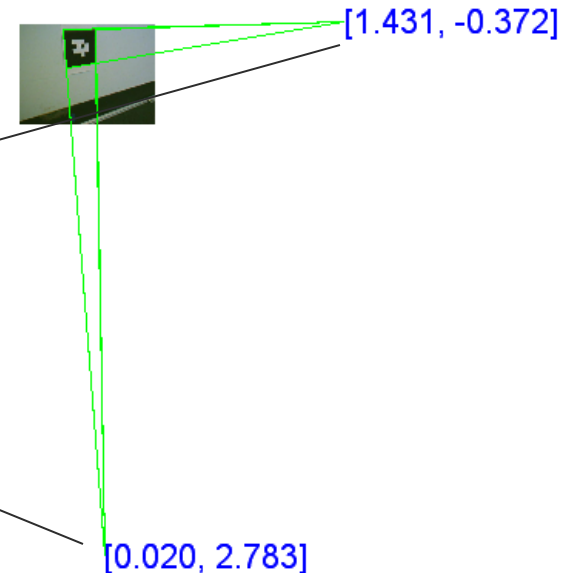
$$r_2 = x_{vY} / \|x_{vY}\|$$

vanishing points of x,y direction,

give us the, $\mathbf{r}_1$ and $\mathbf{r}_2$

$$\begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 & \mathbf{t} \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

first two columns of R



Recovering R from Two vanishing Point (4)

4. Get r_3

$$r_1 = (0.8017, -0.2086, 0.5602)^T$$

$$r_2 = (0.0067, 0.9411, 0.3382)^T$$

$$r_3 = r_1 \times r_2 = (-0.5988, -0.2673, 0.7558)^T \longleftarrow \mathbf{r}_3 = \mathbf{r}_1 \times \mathbf{r}_2$$

5. Get R

$$R = \begin{pmatrix} 0.8017 & 0.0067 & -0.5977 \\ -0.2086 & 0.9411 & -0.2673 \\ 0.5602 & 0.3382 & 0.7558 \end{pmatrix}$$

$\uparrow \qquad \qquad \uparrow \qquad \qquad \uparrow$

$\mathbf{r}_1 \qquad \qquad \mathbf{r}_2 \qquad \qquad \mathbf{r}_3$

Example Recover Camera Pan /Tilt From r_3

$$\alpha = \tan^{-1}(\mathbf{r}_3(1)/\mathbf{r}_3(3))$$

$$\beta = \sin^{-1}(\mathbf{r}_3(2))$$

$$r_3 = (-0.5988, -0.2673, 0.7558)^T$$

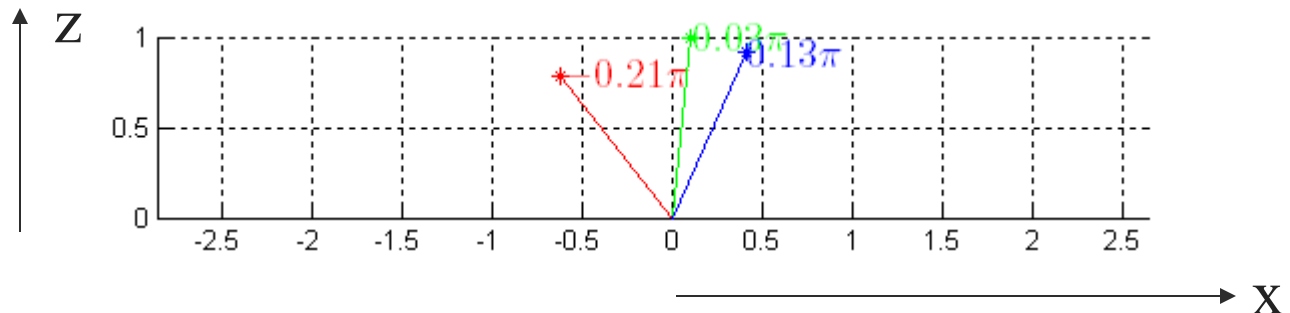
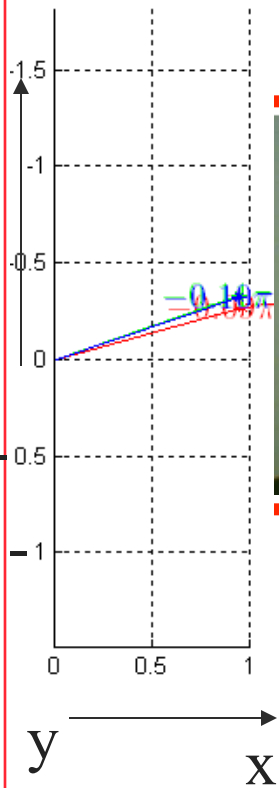
$$\alpha = -0.6691 = -0.2130\pi$$

$$\beta = +0.2706 = +0.0861\pi$$

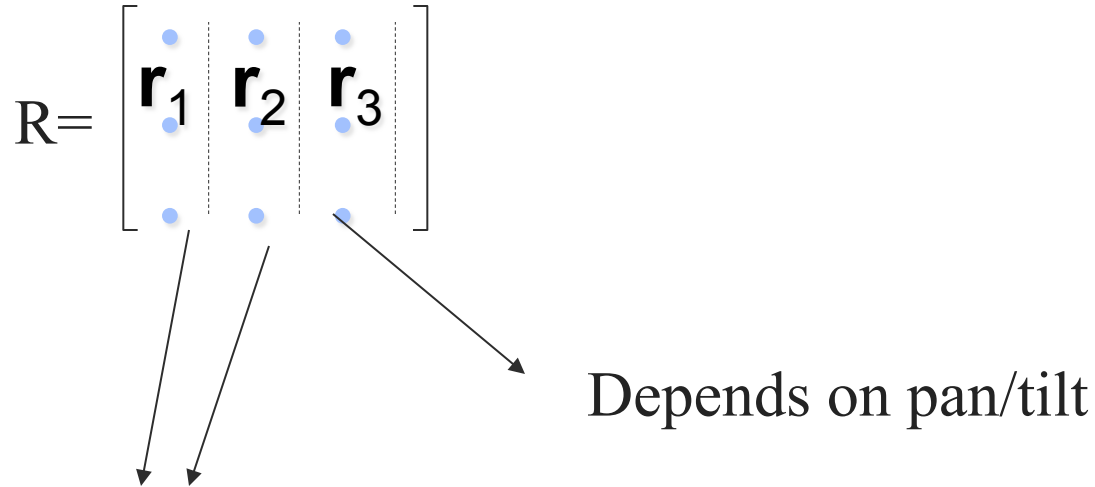


Recover Camera Pan /Tilt From r_3 (2)

Examples of Camera Pan/Tilt angle



Recovering all three angles: yaw + pan/tilt

$$R = \begin{bmatrix} \bullet & \bullet & \bullet \\ \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 \\ \bullet & \bullet & \bullet \end{bmatrix}$$


The diagram shows a 3x3 rotation matrix R enclosed in large square brackets. The middle row contains three vectors labeled $\mathbf{r}_1$, $\mathbf{r}_2$, and $\mathbf{r}_3$. Each vector is represented by a blue dot at the top, the vector label in the middle, and another blue dot at the bottom. Vertical dashed lines separate the columns. Three arrows point from the bottom dots to explanatory text: two arrows from the bottom dots of $\mathbf{r}_1$ and $\mathbf{r}_2$ point to the text "Depends on yaw/pan/tilt", and one arrow from the bottom dot of $\mathbf{r}_3$ points to the text "Depends on pan/tilt".

Depends on yaw/pan/tilt

- 1) Write out equation in yaw/pan/tilt angles for $\mathbf{r}_1$ $\mathbf{r}_2$
- 2) Solve for yaw angle given pan/tilt angles

Summary so far

- 1) If we have the vanishing point in z-direction, we can recover pan/tilt angle
 - 1) This is when the planar target is on the ground
- 2) If don't see the z-direction vanishing point directly, you would need two vanishing points of x-y directions
- 3) Given x-y directions vanishing point, we can cover the full rotation matrix, therefore its z-direction (pan/tilt)
- 4) Given pan/tilt + x-direction vanishing point, we can recover yaw angle

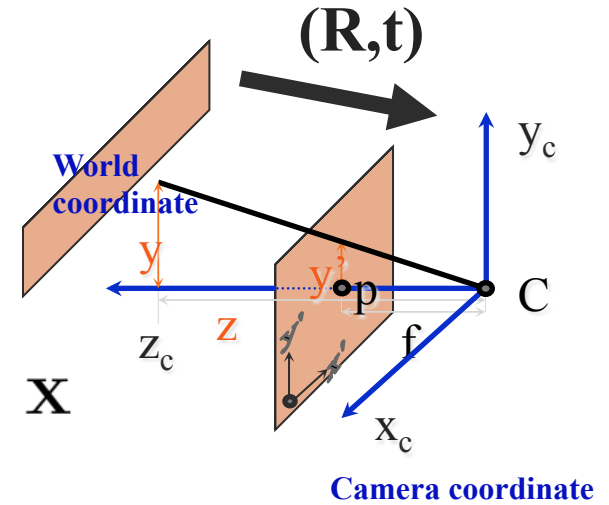
Recover Camera orientation + position

Case 3: homography matrix

Recall for planar surface

$$\mathbf{x} = \mathbf{K} \begin{pmatrix} \mathbf{r1} & \mathbf{r2} & \mathbf{t} \end{pmatrix} \begin{pmatrix} \mathbf{x}_w \\ \mathbf{y}_w \\ 1 \end{pmatrix}$$

$\uparrow$
 $\mathbf{H}_{3 \times 3}$



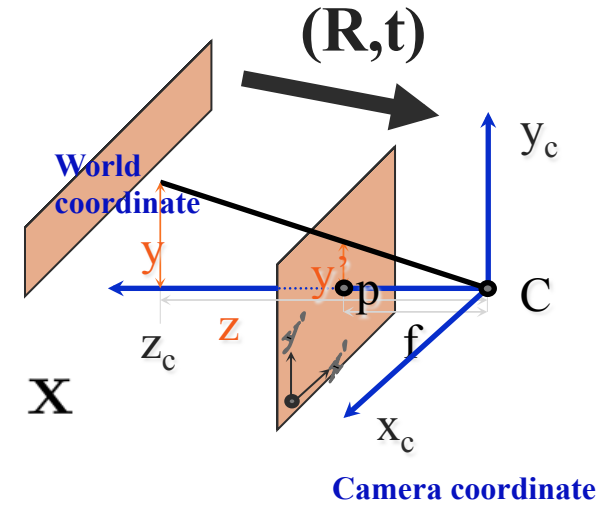
We have the homographic mapping:

$$\mathbf{x} = \mathbf{K} \mathbf{H}_{3 \times 3} \mathbf{X}$$

Recall for planar surface

$$\mathbf{x} = \mathbf{K} \begin{pmatrix} \mathbf{r1} & \mathbf{r2} & t \end{pmatrix} \begin{pmatrix} \mathbf{x}_w \\ \mathbf{y}_w \\ 1 \end{pmatrix}$$

$H_{3 \times 3}$



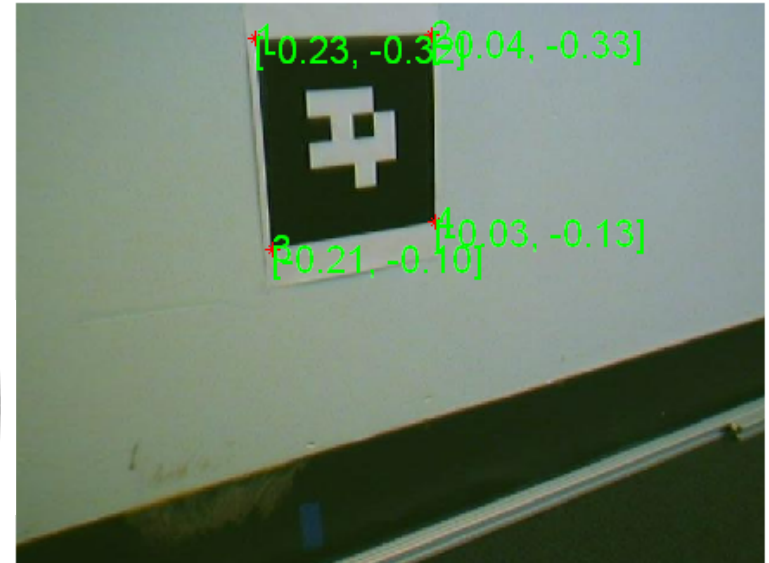
This implies if we have H,

- 1) We can recover the full rotation matrix, R,
- 2) We can recover the position vector t

Example Recover $[R,t]$ from H (1)

1. Get H from 4 points

$$H = \begin{pmatrix} 0.4430 & 0.0037 & -0.1071 \\ -0.1153 & 0.5216 & 0.1506 \\ 0.3096 & 0.1875 & 0.5944 \end{pmatrix}$$



2. Compute the norm of first column

$$a = \|(H_{11}, H_{21}, H_{31})\|$$

Example Recover $[R,t]$ from H (2)

3. Compute r_1, r_2, t

$$\begin{aligned}t &= H(:, 3)/a = (-0.1937, 0.2726, 1.0756)^T \\r_1 &= H(:, 1)/a = (0.8017, -0.2086, 0.5602)^T \\r_2 &= H(:, 2)/a = (0.0067, 0.9439, 0.3392)^T\end{aligned}$$

4. Compute r_3

$$r_3 = r_1 \times r_2 = (-0.1937, 0.2726, 1.0756)^T$$

Procedure of recovery R, and t from H

1) Compute the rotation of x-axis, set

$$\begin{aligned} \mathbf{r}_1 &= H(:, 1); \\ a &= \text{norm}(\mathbf{r}_1) \end{aligned} \quad \mathbf{X} = \mathbf{K} \left(\begin{array}{c|c|c} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{t} \end{array} \right) \begin{bmatrix} \mathbf{x}_w \\ \mathbf{y}_w \\ 1 \end{bmatrix}$$

2) Compute the rotation of the y-axis, set

$$\mathbf{r}_2 = H(:, 2)/a;$$

3) Compute the translation vector t (the position of the robot)

$$\mathbf{t} = H(:, 3)/a;$$

4) Compute the rotation of the z-axis, set

$$\mathbf{r}_3 = \mathbf{r}_1 \times \mathbf{r}_2 \quad R = [\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3]$$

5) Use the previous case(2), to compute the pan/tilt/yaw angle

Summary so far

- 1) Each vanishing point in the image gives you one rotation vector
- 2) Given the z-vanishing point, we can detect the pan/tilt angle.
 - 1) This is the case when the planar target is on the ground.
- 3) Given x-y vanishing point, we can compute the z-vanishing point(therefore pan/tilt angle).
 - 1) This is the case when we are facing the planar surface
 - 2) Given x-rotation vector, pan-tilt angle, we can determine the yaw angle
 - 3) But we can't determine the position of the robot
- 4) If we are given the homograph, (using 4 corresponding points), we can recover both the rotational angles, and the position of the robot.

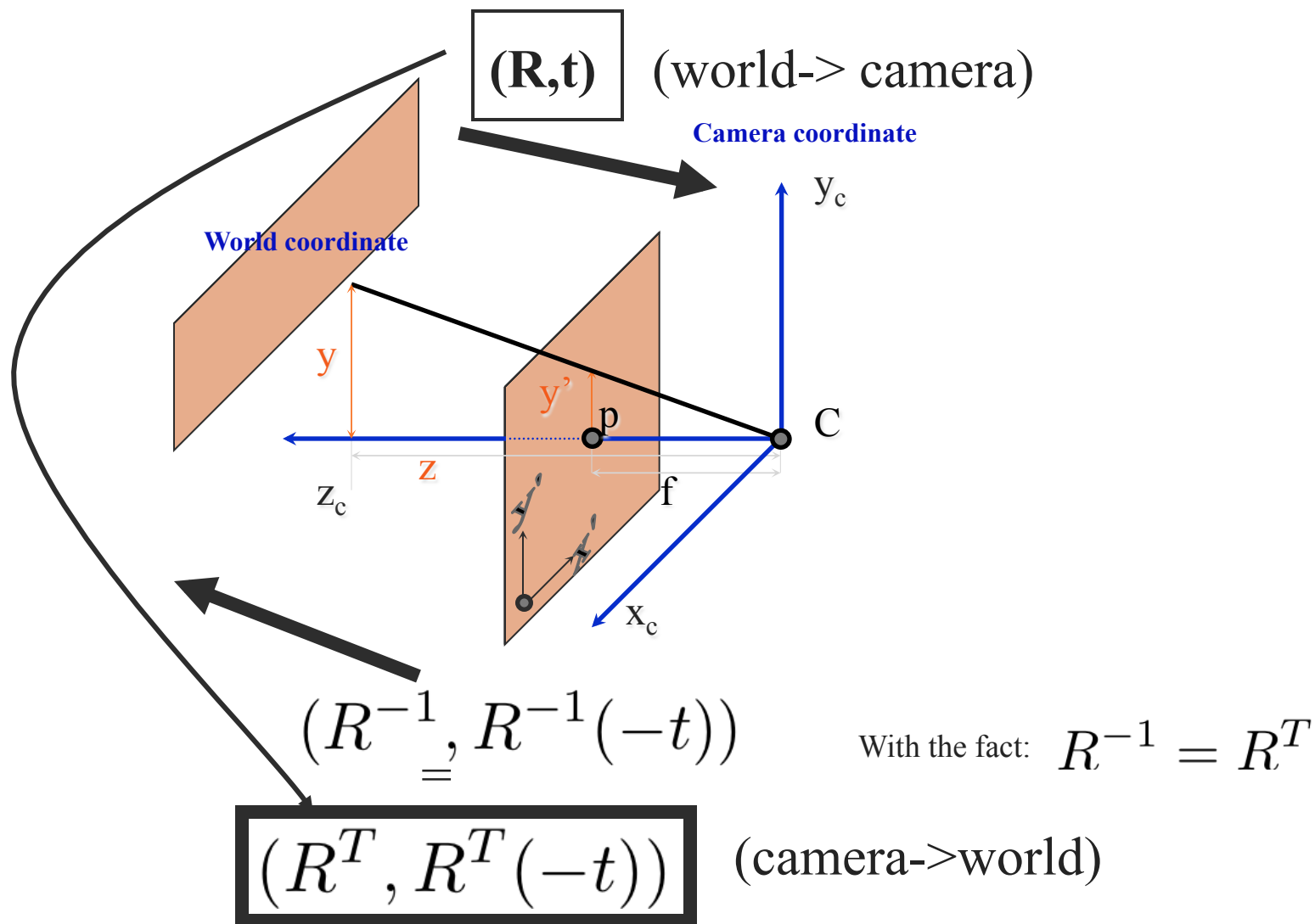
How to estimate the rotation and translation of the robot from the world point of view?

In the case of moving robot (rather than moving target), we need to know the orientation/position of the robot in the world ==>

we need to know how to pan/tilt the world oriented to the robot.

Note: pan/tilt of the camera is very different from the pan/tilt of the world!

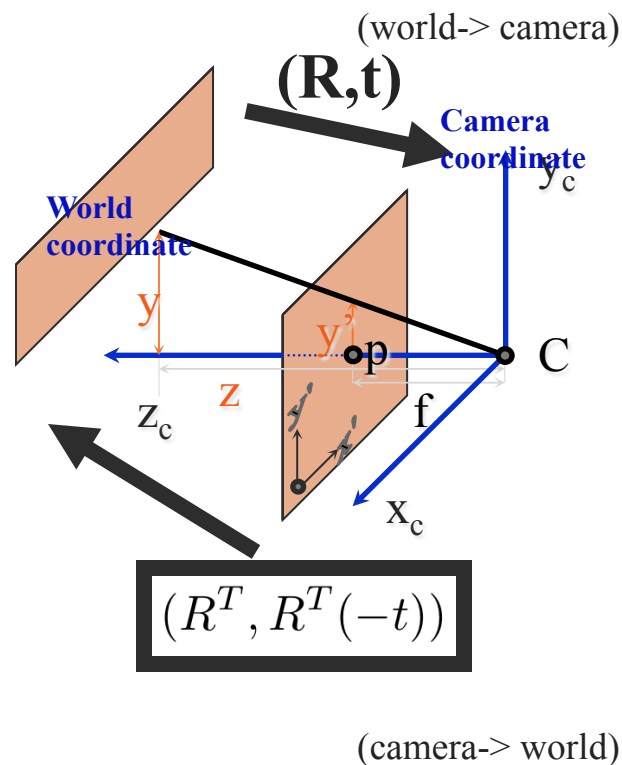
Converting rotation/translation from (world->camera) to (camera->world)



Converting rotation/translation from (world->camera) to (camera->world)

What does it mean:

- 1) The position of the camera in the world reference frame is:
 $R^T t$
- 2) The pan/tilt/yaw of the world to align with camera is computed from the **rows** of the R instead of columns of $\mathbf{R}$



Recovering World pan/tilt from Rotation matrix $\mathbf{R}^T$

1. Compute the (world->camera) rotation $\mathbf{R}$ and translation $\mathbf{t}$
2. Take the last **row** of the $\mathbf{R}$, and recover pan/tilt angle of camera

$$\mathbf{R} = \begin{pmatrix} 0.8017 & 0.0067 & -0.5977 \\ -0.2086 & 0.9411 & -0.2673 \\ 0.5602 & 0.3382 & 0.7558 \end{pmatrix}$$

$$\alpha_{camera} = \tan^{-1}(\mathbf{R}(3, 1)/\mathbf{R}(3, 3))$$

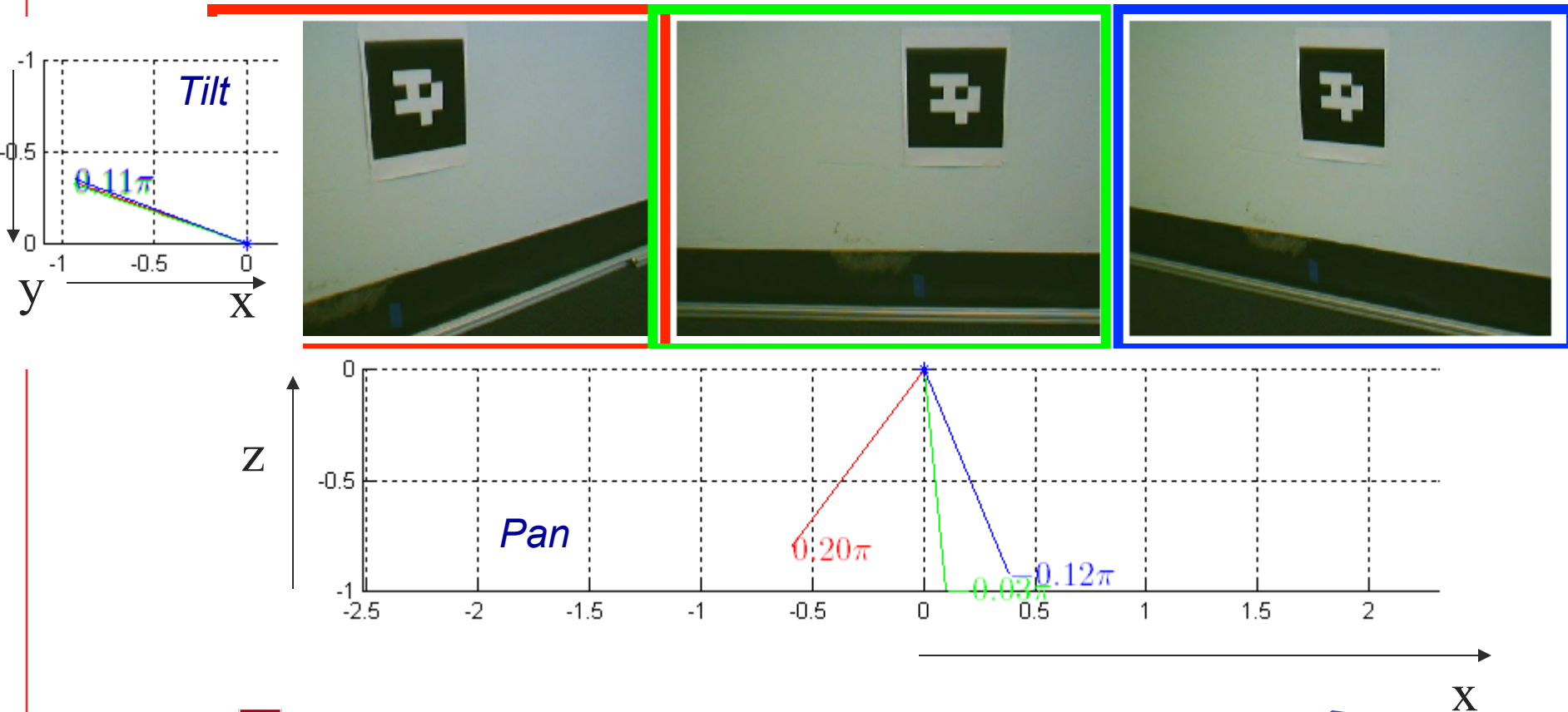
$$\beta_{camera} = \sin^{-1}(\mathbf{R}(3, 2))$$

- 3) Compute the yaw angle if needed, from first two rows of $\mathbf{R}$
- 4) Compute the position of the robot, by $\mathbf{t}_{camera} = \mathbf{R}^T(-\mathbf{t})$

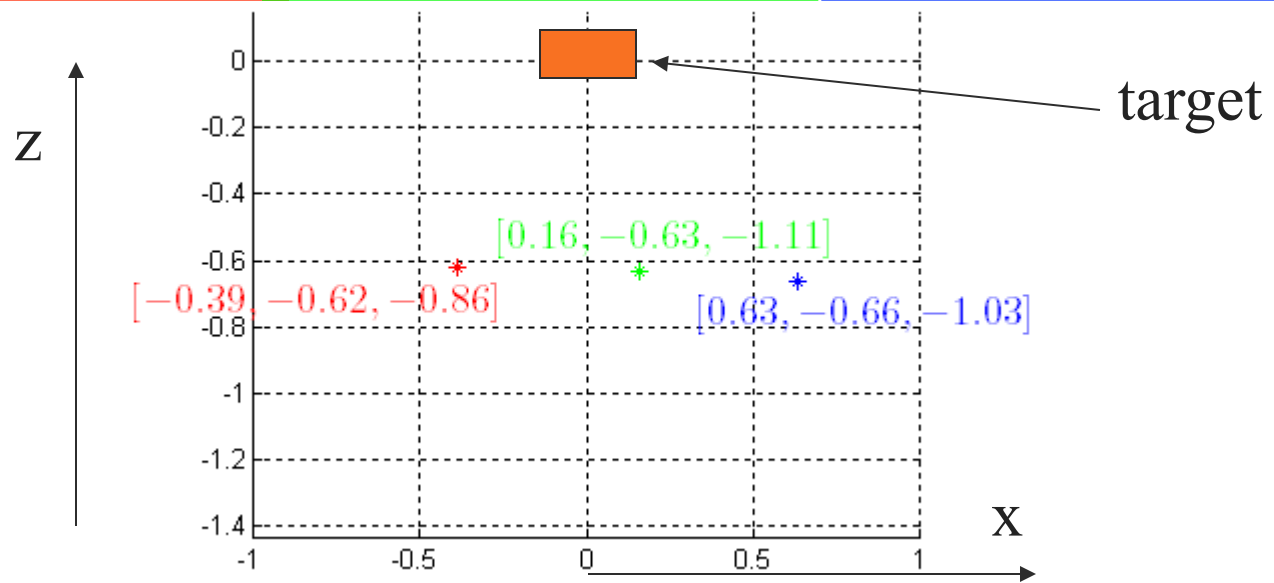
Example Recover Pan/Tilt in *World* Frame

From World $\rightarrow$ Camera: $[R, t]$

$\rightarrow$ From Camera $\rightarrow$ World : $[R^T, -R^T t]$



Example Recover Translation in *World* Frame (2)



Final notes

- 1) There are two rotation/translation we care about:
 - 1) $(\mathbf{R}, \mathbf{t})$: world- $\rightarrow$ camera transformation. It tells us parameters of the camera. The columns of $\mathbf{R}$: rotation of camera so it aligns with the world. The vector $\mathbf{t}$ = coordinate of the world center in camera frame.
 - 2) $(\mathbf{R}^T, \mathbf{R}^T(-\mathbf{t}))$: camera- $\rightarrow$ world. It tells us parameters of the world. Columns of $\mathbf{R}^T$: rotation of the world so it aligns with the camera. The vector $\mathbf{R}^T(-\mathbf{t})$ = coordinate of the camera in the world frame.
 - 3) For the visual odometry task used in project 2, you need to use the transformation $(\mathbf{R}^T, \mathbf{R}^T(-\mathbf{t}))$.
 - 4) The pan angle of the world, is the robot orientation in the world frame.